INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II, Third Semester Examination, October – 2025

Subject Code: BCACC301

Subject Name: Design and Analysis of Algorithms

Full Marks: 30 Duration: 1 Hour

Part – A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. Suppose you have two algorithms for the same problem: one uses divide and conquer, and the other uses dynamic programming. Under what conditions would dynamic programming be preferred? **2**
2. Why do we analyse space complexity if memory is cheaper nowadays? **2**
3. If algorithm A takes O(nlogn) and algorithm B takes $O(n^2)$, for what values of n might algorithm B still be faster? Why? **2**
4. What is the difference between divide and conquer and dynamic programming? **2**
5. Consider an algorithm with time complexity $O(2^n)$. Can it ever be practical? Under what circumstances? **2**
6. Why is NP-completeness important in algorithm design? **2**

Part – B
Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. Compare BFS and DFS in terms of time complexity, space complexity, and applications. In which situation would you prefer BFS over DFS and why? **5**
8. The 8-Queen problem is often solved using backtracking. Discuss how backtracking helps reduce the search space compared to a brute-force approach. **5**
9. Differentiate between NP-hard and NP-complete problems with proper examples. Why is proving a problem NP-complete important in computer science? **5**

Part – C

Answer any 1 question out of 2

Each question carries 10 marks (10 × 1)

10. TSP is an NP-hard problem. Justify why heuristic approaches (like nearest neighbour or branch and bound) are often used instead of exact algorithms for large inputs. **10**

11. Construct a Huffman tree and find the average code length for the following set of characters with their frequencies: **10**

Character	Frequency
A	5
B	9
C	12
D	13
E	16
F	45

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